



# Multi-Arm Multi-Stage Trials (MAMS)

## *A Practical Introduction to Simulating Complex Trial Designs*

Thomas Jaki<sup>1,2</sup> & Dominique-Laurent Couturier<sup>2</sup>

<sup>1</sup> University of Regensburg, DE

<sup>2</sup> MRC Biostatistics Unit, University of Cambridge, UK

<https://github.com/dlc48/CEN2026>

# Defining a multi-stage trial

We now consider **splitting the trial over  $J$  stages**

In the first stage of the trial, patients are recruited to all groups, i.e.,  $T_0, \dots, T_K$

After each trial stage  $j = 1, \dots, J$ , treatment arms may be dropped either for futility or for efficacy.

If the trial is not stopped at an interim analysis, we recruit patients for the shared control group and remaining treatment arms at the next stage.

# Sampling scheme in a 2-stage case

In stage 1, we recruit patients to all of the treatments



We then conduct an [interim analysis](#) to choose which treatments to continue recruiting from



This may be extended to many stages conducting interim analyses between each stage of the trial

# Different types of MAMS designs

Different MAMS designs can be defined based on:

- ▶ stopping rules
  - ▷ per arm
    - early futility stopping or not
    - early efficacy stopping or not
  - ▷ for the trial
    - simultaneous stop:  
trial stops as soon as at least one treatment is found efficacious
    - separate stop:  
trial continues as long as at least one treatment is 'promising'
- ▶ on the maximum number of arms allowing to reach the  $j$ th stage
  - ▷ unlimited
  - ▷ only 1

## Two-stage selection screens

Suppose that responses to the  $k$ th treatment,

$$Y_{ki} \sim N(\mu_k, \sigma^2), \quad k = 0, 1, \dots, K$$

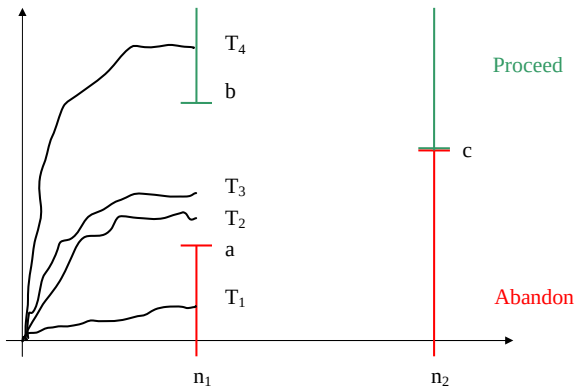
In Stage 1,  $n_{1k}$  patients are treated on  $T_k$ , with mean response  $\bar{y}_{1k}$  with test statistic (where  $n_{jk}$  denote the sample size of  $T_k$  at stage  $j$ )

$$Z_{1k} = \frac{\bar{y}_{1k} - \bar{y}_{10}}{\sqrt{\frac{\sigma^2}{n_{1k}} + \frac{\sigma^2}{n_{10}}}}$$

After Stage 1, an interim analysis is conducted

- ▶ If  $Z_{1k} < f_1$ , then  $T_k$  will be dropped
- ▶ If  $Z_{1k^*} = \max(Z_{11}, \dots, Z_{1K}) > e_1$ , then  $T_{k^*}$  will be selected

## Early stopping : Visualisation



- ▶  $T_1$  eliminated and  $T_4$  selected at interim
- ▶ AMEND PLOT TO HAVE  $f_1$ ,  $e_1$  and  $e_2$  INSTEAD OF  $a$ ,  $b$ ,  $c$
- ▶ AMEND PLOT TO HAVE 'Stage 1', 'Stage 2' INSTEAD OF  $n_1$ ,  $n_2$

# Only the best treatment moved forward to stage 2

(e.g. Stallard & Todd, 2003; Whitehead & Jaki 2009)

If no treatment is selected after Stage 1, and not all treatments are dropped, then

- ▶  $n_{20}$  further patients are treated on the control
- ▶  $n_{2k^*}$  on treatment  $T_{k^*}$ , the best remaining active treatment

At the end of Stage 2, the statistic of the remaining treatment is given by

$$Z_{2k^*} = \frac{\bar{y}_{2k^*} - \bar{y}_{20}}{\sqrt{\frac{\sigma^2}{n_{.k^*}} + \frac{\sigma^2}{n_{.0}}}},$$

where  $\bar{y}_{2k^*}$  denotes the mean response in group  $k^*$  defined over the  $n_{.k^*} = \sum_{j=1}^2 n_{jk^*}$  patients.

If  $Z_{2k^*} > e_2$ , then  $T_{k^*}$  will be selected (declared efficacious).

# All promising treatments moved forward to stage 2

(Magirr et al, 2012)

If no treatment is selected after Stage 1, and not all treatments are dropped, then

- ▶  $n_{20}$  further patients are treated on the control
- ▶  $n_{2k}$  on treatment  $T_k$  for all  $k$  so that  $Z_{1k} > f_1$

At the end of Stage 2, the statistics of the  $k$ th treatment is given by

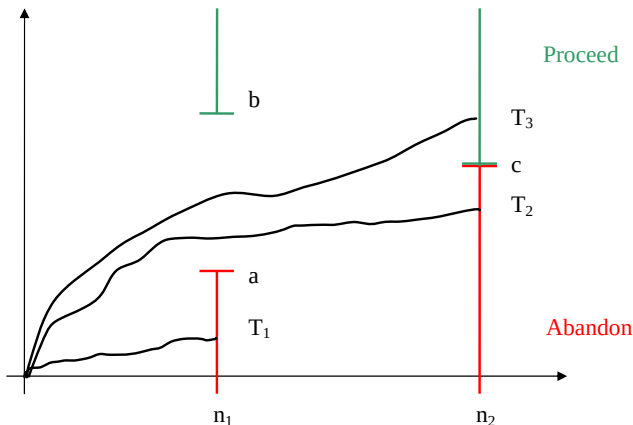
$$Z_{2k} = \frac{\bar{y}_{2k} - \bar{y}_{20}}{\sqrt{\frac{\sigma^2}{n_{\cdot k}} + \frac{\sigma^2}{n_{\cdot 0}}}},$$

where  $\bar{y}_{2k}$  denotes the mean response in group  $k$  defined over the  $n_{\cdot k} = \sum_{j=1}^2 n_{jk}$  patients.

If  $Z_{2k^*} = \max(Z_{21}, \dots, Z_{2K}) > e_2$ , then  $T_{k^*}$  will be selected.



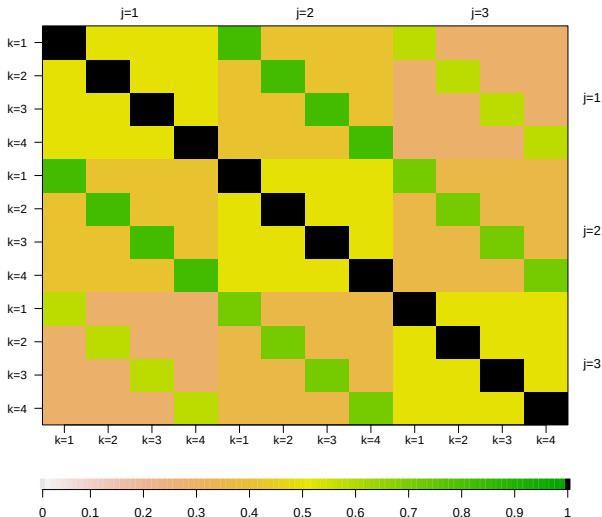
## Conclusion at stage 2: Visualisation



- ▶  $T_1$  eliminated at interim and  $T_3$  selected at final analysis
- ▶ AMEND PLOT TO HAVE  $f_1$ ,  $e_1$  and  $e_2$  INSTEAD OF  $a$ ,  $b$ ,  $c$
- ▶ AMEND PLOT TO HAVE 'Stage 1', 'Stage 2' INSTEAD OF  $n_1$ ,  $n_2$

# Within/between 'Stage and treatment arm' dependence

ADD HERE FORMULA FOR  $Cov(Z_{jk}, Z_{lm})$  and EXPLAIN SETTING ( $r$ ,  $r_0$ )



## Constraints for finding a design

Under the global null:

$$\nu = P(\text{select a treatment} \mid \mu_0 = \mu_1 = \cdots = \mu_K)$$

Under the alternative:

$$1 - \eta = P(\text{select } T_1 \mid \mu_1 = \mu_0 + \delta \text{ and} \\ \mu'_{k'} = \mu_0 + \delta_{0}, \quad k' = 2, \dots, K - 1).$$

► Given the allocation ratios  $r_k = n_{\cdot k} / n_{\cdot 0} \Rightarrow$  find  $f_1, e_1, e_2$  and  $n_{0j}$

► Further constraints can be set, such as

- (i)  $f_1 = 0, e_1 = \infty$
- (ii)  $f_1 = 0, e_1 = \frac{4}{3}e_2$
- (iii)  $f_1 = 0$  and  $e_1, e_2$  defined via O'Brien-Fleming rule

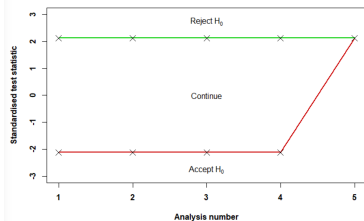
and a solution can be computed and evaluated

## Efficacy and futility bounds:

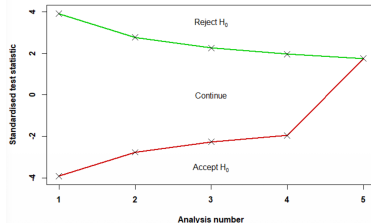
- ▶ The shape of the stopping boundaries will determine the ESS and MSS.
- ▶ Two main ways to choose between the infinite number of possible shapes:
  - ▷ Use some 'fixed' boundary shape (specified through a simple functional form)
  - ▷ Search for an optimal design to minimise a criterion for some  $\delta$  (ESS).
- ▶ First method is much quicker, but second method allows greater control over the ESS properties of the design.
- ▶ Common boundary shapes to choose from:
  - ▷ Pocock
  - ▷ O'Brien-Fleming
  - ▷ Triangular test

# Efficacy and futility bounds: Visualisation

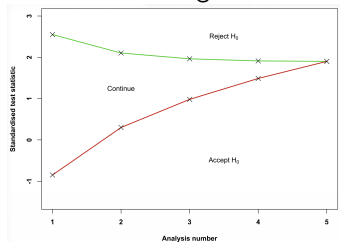
Pocock



O'Brien & Fleming



Triangular



## Application – the `MAMS::mams()` function

The `mams()` function of the MAMS package allows to design studies with asymptotically normal endpoints and known variance. It considers

- ▶ normal, binary, ordinal and time-to-event endpoints
- ▶ different arm/trial stopping rules
  - ▷ `method="simultaneous"` (default):
    - all promising arms of stage  $j - 1$  are moved forward to stage  $j \leq J$ ,
    - both efficacy and/or futility early stopping are possible,
    - the trial stops as soon as one treatment is found efficacious.
  - ▷ `method="dtl"` (drop-the loser):
    - at each stage, a pre-defined number of treatment arms if moved to the next stage
    - neither efficacy and futility early stopping are possible,
    - efficacy can only be claim for the only treatment considered at stage  $J$
  - ▷ `method="sep"` (separate):
    - all promising arms of stage  $j - 1$  are moved forward to stage  $j \leq J$ ,
    - both efficacy and/or futility early stopping are possible,
    - the trial keeps going as long as at least one 'promising' treatment remains.

# Application – the trial TREADON

Let us consider a variant of the TREADON trial with two stages and two treatment arms and let's assume that

- ▶ assumed effect sizes and uncertainty:  $\delta = 0.8$ ,  $\delta_0 = 0$ ,  $\sigma = 2.7$
- ▶ target type I and type II errors:  $\nu = 0.05$ ,  $1 - \eta = 0.9$
- ▶ allocation ratios: same sample size for each group at each stage
- ▶ efficacy stopping rules: O'Brien-Fleming
- ▶ futility stopping rules: fixed value equal to 0

```
> library(MAMS)
> design1 = mams(K=2, J=2, alpha=0.05, power=0.9, r=1:2, r0=1:2,
  delta=0.8, delta0=0, sd=2.7,
  ushape="obf", lshape="fixed", lfix=0,
  method="simultaneous",
  type="normal", parallel=TRUE)
```

## Application – the trial TREADON

Some results ('simultaneous' stopping rule)

| $f_1$ | $e_1$           | $e_2$ | $n_{10}$ | $N$ | $E_0(N)$ | $E_{LFC}(N)$ |
|-------|-----------------|-------|----------|-----|----------|--------------|
| 0     | $\infty$        | 1.910 | 118      | 708 | 550      | 646          |
| 0     | $-\frac{4c}{3}$ | 1.945 | 120      | 720 | 556      | 532          |
| 0     | 2.731           | 1.931 | 119      | 714 | 555      | 546          |

$N$  is the maximum sample size

$E_0(N)$  and  $E_{LFC}(N)$  = expected total sample size under the null and least favourable configuration respectively.



## Where simulation can help

- ▶ Estimating the ESS (practical 2, task 4)
- ▶ Investigating the effect of drop-out scenarios (practical 2, task 5)
- ▶ Estimating the probability of reversed conclusions (practical 2, task 6)
- ▶ Estimating biases related to early/late stopping (practical 2, task 7)
- ▶ Estimating the duration of the study (practical 2, task 8)
- ▶ Delay in observing endpoint/overrun (practical 2, task 8)
- ▶ ...

# References

- ▶ Dunnett, C. (1955). A Multiple Comparison Procedure for Comparing Several Treatments with a Control. *Journal of the American Statistical Association*, **50(272)**, 1096-1121.
- ▶ Magirr D, Jaki T, Whitehead J (2012) A generalized Dunnett test for multi-arm, multi-stage clinical studies with treatment selection. *Biometrika*. **99(2)**:494-501.
- ▶ Stallard N and Todd S (2003) Sequential Designs for Phase III Clinical Trials Incorporating Treatment Selection. *Statistics in Medicine*. **22**:689-703.